



INTERNATIONAL YEAR OF
**ARTISANAL FISHERIES
AND AQUACULTURE**
2022

“Fishing for Life 2022”

**South and South-East Asian
Conference on
Small Scale Fisheries and Aquaculture
19th and 20th September 2022**



BOOK OF ABSTRACTS

**Sri Lanka Forum for Small Scale Fisheries
(SLFSSF)**



Proceedings of the South and South-East Asian Conference on Small Scale Fisheries and Aquaculture: 2022

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Fishing for life-2022

South and South-East Asian Conference on Small Scale Fisheries and Aquaculture



**South and South-East Asian Conference on Small Scale
Fisheries and Aquaculture**
19th -20th September 2022

“Fishing for Life 2022”

Book of Abstracts

Thematic Areas

Sustainable use of resources

Impact of Policy, Technology, Trade and Development on SSF

Poverty, Rights and Rural Aquaculture

Gender issue in SSF

Governance and Management of SSF and Implementation of SSF guidelines

Sri Lanka Forum for Small Scale Fisheries (SLFSSF)





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Aims of ‘Fishing for Life Conference’: SACSFA 2022

Emeritus Professor Oscar Amarasinghe

PRESIDENT / Sri Lanka Forum for Small Scale Fisheries (SLFSSF)



Our conference is all about people and their families engaged in small-scale fisheries, a sector that makes an important contribution to employment, nutrition, food security, sustainable livelihoods and poverty alleviation – especially in developing countries. As, the FAO notes, “despite this significant contribution, the issues constraining the sustainable development of small-scale fisheries remain poorly understood” which is true for the south and Southeast Asian region, as well.

High rates of resource exploitation, unregulated technological change, increasing fishing pressure, pollution of both coastal and inland waters, climate change impacts, etc. are causing threats to aquatic ecosystems, leading to resource degradation. While facing declining catches and income, small-scale fishers have also been excluded from managing their own resources. Both their fishing rights and human rights have been violated. Recent trends in Blue Economic Growth initiatives have revealed, apart from their benefits to various stakeholders, a process of marginalization of small-scale fishing communities. There is widespread poverty among SSF communities and the livelihoods of many have become non-sustainable. It is thus evident that it will be quite difficult to achieve the dual management objectives: resource sustainability and human wellbeing. How we are going to resolve it consists of the governance challenge today.

It is in this particular context that the Sri Lanka Forum for Small Scale Fisheries (or SLFSSF) was formed in February 2018 to address the major issues preventing the attainment of a sustainable small-scale fisheries sector. This forum is a partnership or Unity, of academics, researchers, state officers, community organisations, and civil society organisations. It is Unity in Diversity.

With the declaration of the year, 2022 as the Year of Artisanal Fisheries and Aquaculture (IYAFA), the SLFSSF decided to organize an event to highlight various issues threatening the SSF communities to draw the attention of policymakers and planners to these issues as a means of securing a sustainable small scale fisheries sector. This is the birth of the Fishing for Life Conference, or South and Southeast Asian



Conference on Small Scale Fisheries and Aquaculture. The term “Fishing for Life” was chosen to show that fishing is not simply “catching fish, selling fish and earning an income”, but a particular way of life; values, beliefs, culture and legal system, which is deeply rooted in small scale fishing communities. We also felt that issues pertaining to small-scale fisheries are common to all our countries in the south and Southeast Asian region. Therefore, we thought that we would organize a conference allowing all countries in the region to present their cases so that we can finally understand issues common to the region and put all our brains together to find the most appropriate governance and management mechanisms to manage the SSF sector.

SACSFA 2022 will be a unique conference because it is allocating equal weight to academic sessions and video presentations by Civil Society Organisations, in the region which is termed the “Voice of the People”. Thus, we will have the opportunity to listen to various research done in the region and their policy implications while listening to what fishing people in the region have to say about their own lives; what issues they confront, and what needs to be done to resolve them. We also intend to make this event a biennial conference, that will bring into focus incessantly all issues that are interwoven with the fishing way of life.

SACSFA 2022 is a two-day conference. We will have 7 keynote speeches, made by world-renown scientists: Emeritus Professor Maarten Bavinck from the University of Amsterdam, in the Netherlands, and Emeritus Professor Svein Jentoft from the Arctic University in Norway. Professor Anthony Charles from Saint Mary's University in Canada, Professor Meryl J. Williams from Gender in Aquaculture and Fisheries Section of the Asian Fisheries Society, Mme Franz Nicole, from the Food and Agriculture Organisation of the United Nations, Dr. Premachandra Wattage from the Sabaragamuwa University of Sri Lanka, and Emeritus Professor Oscar Amarasinghe from the University of Ruhuna, Sri Lanka.

Technical sessions are divided into 5 major themes: Sustainable Use of Resources, Impact of Policy, Technology, Trade and Development on SSF, Poverty, Rights and Rural Aquaculture, Gender Issues & Governance, and Management of SSF. We have about 40 scientific papers to be presented and 10 video presentations by Civil Society Organisations. Each day, the time allocated for all sessions would roughly be about 5 hours.

At the end of the second day, the stage will be prepared for a final panel who would, by taking into account various policy implications, recommendations, and guidelines emerging from the conference, prepare a declaration: “SACSFA 2022 Declaration”. The panel includes Mr. Sebastian Mathew, the executive director of the International





Collective in Support of Fish workers (ICSF), Prof. M. J. Williams, Mme Nicole Franz, Prof. M. G. Kularatne, Professor of Economics, University of Kelaniya, Sri Lanka, Dr. Krihnan Pandian, Director of the Bay of Bengal Programme, and Mr. Herman Kumara, the Convener of the National Fisheries Solidarity, Sri Lanka. This will be the final outcome of the whole exercise that will show us the way forward in reaching the dual objective of sustainable resource management and human wellbeing.

My sincere wishes to all for a very interesting, exciting and productive conference that will contribute, along with other IYAFA initiatives, towards securing sustainable small-scale fisheries and aquaculture.



Keynote Address (Summary)

Emeritus Professor Marrten Bavinck

Moving histories of small-scale fishing peoples

My keynote for this conference bears as main title: 'Moving histories of small-scale fishing peoples'. Each term in this title is of relevance. Of course, the keynote will be about small-scale fishers (SSF), as this is the conference theme and also one of my main research interests. It is also about 'histories' as constituting the ongoing passage of time. Finally, it is about 'moving', in the sense of 'heart-rending', but also of 'change' in a spatial and a temporal sense. My argument will be that the SSF sector is not a fixed, but a changing entity. It is embedded in broader societal structures and processes. Secondly, the people populating SSF make continuous livelihood choices that take them to different places and positions within SSF, as well as outside. I will illustrate my argument with a history from northern Sri Lanka, where I carried out my first social-science fieldwork in 1977 and have returned ever since. It focuses on the fishing village of Kadalur and on four generations of SSF which started with fish vendor Kannama. Obviously, this history will be of up's and down's.



Prof. Marrten Bavinck
Emeritus Professor
University of Amsterdam
Netherlands

Keynote Address (Summary)

Emeritus Professor Oscar Amarasinghe

Rights, Poverty and Sustainable Small-Scale Fisheries

Poverty in fisheries, especially in the marine domain, is popularly considered a result of the common property nature of fisheries resources, a manifestation of Garret Hardin's Tragedy of the Commons. This old paradigm of poverty is often related to income. How poverty was perceived by various persons has changed over time, and thus the approaches to poverty alleviation and development. They changed from increased income to favourable distribution of income, to basic needs, gender equity and equality, human development, sustainable livelihoods, etc., and today there is increased concern on human rights, culture, institutions, access to basic social services, etc. In his seminal article on "when fishing rhymes with poverty" Chris Bene (2003) mentions two contrasting interpretations of the relationship between poverty and fisheries; "they are fishermen because they are poor" and "they are poor because they are fishermen". The first relates to the role of fisheries as last resort employment for the poor, while the latter is related to conventional wisdom and overexploitation of resources due to the open access nature of fisheries, low opportunity costs, etc. Both perspectives are important in understanding poverty in fisheries.

In recent years, the focus has shifted to the rights of the fishers, both rights of access to resources and human rights. Today, a "rights-based" approach to development is emerging as an overriding development strategy that argues that human rights are integral to human development. Profit maximization and economic efficiency are considered as insufficient goals, because they ignore the issues of social responsibility, distributive justice, rights to food and basic social services, etc. Along with the rights of fishers, governance and management failures, blue injustices, social development issues, marginalization of SSF communities, etc., have come to the policymaking and planning table as issues needing immediate attention. These issues, together with the strong nature-dependency of fisheries, the hazardous environment in which fishers operate, the uncertainty of fish catches, as well as the risky nature of many fishing operations have made the fishers a highly vulnerable group. More than resource access issues alone, it is fishers' inability to cope with the vulnerability that has had a more detrimental impact on fishing livelihoods, pushing them into dumps of poverty.

Today, co-management is proposed as a panacea to fight vulnerability in fisheries, that is expected to allow fishers to voice and exercise their rights; both access and human rights, resolve social development issues, actively participate in decision-





making processes, etc., which will have a powerful positive impact on poverty reduction and securing sustainable livelihoods. However, this requires a strong government will and commitment to make things right, especially to ensure that co-management is a truly interactive platform, characterized by integration, inclusion, and effective participation of all stakeholders with a holistic approach to management. It needs to be emphasized that prescriptions for securing fishers' rights, sustainable livelihoods, social development, co-management, training, and capacity building of diverse stakeholders have all been incorporated into SSF guidelines, which need to be wisely used to remodel fisheries policy and management.

Oscar Amarasinghe Ph.D.

Emeritus Professor / University of Ruhuna, Sri Lanka

Chancellor / Ocean University of Sri Lanka

President / Sri Lanka Forum for Small Scale Fisheries (SLFSSF)



Keynote Address (Summary)

Professor Anthony Charles

Small-Scale Fisheries Manage Resources and Conserve the Environment



The Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries (SSF Guidelines) highlight the importance of rights and responsibilities, as well as the links between them. Respect for the rights of fishers to make their livelihoods from fishing and for human rights connects with the responsibility to sustain resources and ecosystems over time. The fact to be emphasized is that for small-scale fishers, this stewardship responsibility is not something new. In many cases, fishing communities and organizations have long histories of conserving and stewarding their fishery resources and ecosystems, supporting the health and livelihoods of their communities. That reality is acknowledged in the SSF Guidelines, which note that "States should recognize the role of small-scale fishing communities and indigenous peoples to restore, conserve, protect and co-manage local aquatic and coastal ecosystems".

This presentation provides a focus, within the International Year of Artisanal Fisheries and Aquaculture (2022), on environmental conservation and stewardship of small-scale fishers, and how governments and others can best support those actions. The discussion draws on reports contributed by dozens of fisher organizations and communities around the world, leading to an online resource highlighting SSF stewardship activities, the SSF-Stewardship Project (<https://ssf-stewardship.net/>). The stewardship of small-scale fishers is rooted in values, respect for the natural world and biodiversity, and a desire for sustainable livelihoods and food security. As noted by contributor, Muhamad Syahrul of Indonesia, "The sea as a source of livelihood must be maintained and preserved so that it is sustainable for future generations." The stewardship practices of small-scale fishers fit into four major categories, each described in the presentation:

Physical activity: physical efforts such as restoring mangroves, protecting fish habitat, or caring for species such as turtles and seahorses, and dealing with climate change impacts due to, for example, flooding, coastal erosion or saltwater incursion.

Managing fisheries: fishery management measures contributing to environmental stewardship, such as controlling where or when fishing can take place or fishing gear used, and taking part in government-led fishery co-management.

Managing the use of aquatic areas: measures such as closed areas or sacred sites or other conservation tools dealing with specific aquatic areas, and often applying not only to fisheries but also other human uses of the area.

Broader-scale engagement and collaboration: engagement of the fishing

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Sri Lanka Forum for Small Scale Fisheries (SLFSSF)



community or organization in stewardship organized at a level broader than its local area, in initiatives led by governments or others, such as fishery management or conservation projects.

Three more stewardship categories provide support for the above, as important elements of stewardship themselves:

Building knowledge and capacity: collecting scientific data, or utilizing local knowledge, to support stewardship, and directly building or enhancing the stewardship capacity of the community or organization.

Monitoring: keeping track of the state of aquatic habitats, ecosystems, biodiversity, and climate) and of the activities that may impact the environment (e.g., coastal development).

Communications, education and outreach: activities directly and concretely supporting stewardship of oceans or inland waters through education, advocacy, communications, outreach or other related activities.

The presentation highlights how the actions of small-scale fishers to manage resources and conserve the environment are rooted in rights, livelihoods, and values. As a contributor, Duangporn Chumpholwong of Thailand notes, "The unity and participation of the people in the community is an important part of the successful care of resources."

Anthony Charles
Saint Mary's University, Halifax, Canada



Keynote Address (Summary)

Dr. Premachandra Wattage

Impact of Technology, Trade, Development, and Policy (TTDP) on Small Scale Fisheries

Marine-based small-scale fisheries (SSFs) in Asia face an uncertain future due to overexploited fisheries and climate change-related declines projected in tropical fisheries. SSFs provide a significant source of animal protein, income, and employment, accounting for a substantial part of the world's fish production – at least 40 percent – and are particularly important in Asia. However, the contribution of SSFs to national fish supply is rare in regional fisheries management in Asia. As such, SSFs tend to be significantly underestimated compared to industrial fisheries. Further, the lack of knowledge about the functions of the SSFs sector is at risk due to associated problems with technology use, trade, development, and policy. Therefore, informed decision-making for sustainable ecosystem-based fisheries management and social development is at risk for SSFs in the Asian region.



SSF employs millions of men and women fishers directly engaged in fishing activities and millions working in allied activities such as fish catching, processing, marketing, boat building, and net making. With the help of family members, rural Asian people depend on fisheries for their livelihood. In Asia, fish accounts for more than 50 percent of the total animal protein intake. In this region, SSF provides over three-quarters of the domestic fish supply. In southeast Asia, possibly a billion people rely predominantly on fish for animal protein. A fundamental problem of most SSF families in the region is their poor low standard of living and frequent poverty levels. However, the record shows remarkable national economic growth and overall fisheries development. Based on the literature review, this speech attempts to look at how TTDP is not supporting the poor livelihood of Asian fishers. Is it due to the TTDP or any other factors?

SSFs exploit under an open access regime, sometimes enforced by development policies. Even well-organized, traditional social mechanisms may have existed to restrict such access; these continued through the country's overall political system. It has led to excess fishing capacity, resource depletion, waste of economic and human resources, and poor returns on development efforts, combined with the fish demand and supply mechanism in the value chain. The sustainable development of small-



scale fisheries is desirable for balanced social, economic, and regional development in coastal and rural areas. It would require, among other things, addressing development constraints faced by SMFs by creating more development opportunities. Achieving it requires introducing sustainable ecosystem-based fisheries management over traditional fisheries management. Such a system requires regional corporations within neighboring countries of Asia.

Dr. Premachandra Wattage

Consultant, Faculty of Agriculture, Sabaragamuwa University of Sri Lanka.

Visiting Research Fellow, University of Portsmouth, UK



Keynote Address (Summary)

Mme. Franz Nicole

The Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication (SSF Guidelines)

The Voluntary Guidelines for Securing Sustainable Small-Scale Fisheries in the Context of Food Security and Poverty Eradication (SSF Guidelines) have been endorsed by the FAO Committee on FAO in 2014, after a participatory development process that involved over 4,000 stakeholders around the world. Small-scale fisheries organizations were a main driver in this development process. These SSF Guidelines are grounded in a human right-based approach, and they go far beyond the topics usually dealt with in a fisheries instrument. The SSF Guidelines do of course deal with fundamental issues such as access to aquatic resources and sustainable fisheries management. But they also include chapters on social development, employment and decent work; value chain, post-harvest and trade; gender equality; and disaster risks and climate change. They put people at the center, while acknowledging that rights and responsibilities go together.



A balanced and equitable partnership approach and cross-sectoral collaboration is fundamental for the implementation of the SSF Guidelines. There are clearly different roles for different partners and the implementation of the SSF Guidelines follows a dual strategy of “top-down” and “bottom-up”:

- Governments have a key responsibility to make implementation happen at the national and local level, and to incorporate the SSF Guidelines (principles) in relevant policies, strategies, etc. – not only for fisheries but for overall socio-economic development. Political engagement at high level together with investments in capacity and participatory decision-making processes is required in order to realize the new vision for small-scale fisheries.
- Fishers and fish workers, through their organizations, are main drivers of change and play a major role in the “bottom-up” processes. The SSF Guidelines need to empower collective action to ensure that small-scale fisheries are mainstreamed into relevant policies, strategies and actions at the local, national and regional level, and to ensure the implementation of these policies, strategies and actions.
- Other players, in particular academia and research, but also regional organizations, NGOs and others, should be at the interface of this dual strategy, with a function to connect, to supplement, to document and to strengthen the above-mentioned efforts.



Since the endorsement of the SSF Guidelines in 2014 FAO is working with partners at the global, regional and national level to support their implementation, using the above outlined approach. This includes the promotion of the SSF Guidelines in relevant global and regional processes and organizations. In fact, most relevant regional organizations by now refer to the SSF Guidelines in their strategies and initiatives. In addition, FAO has developed a number of thematic guidance documents, e.g. on gender, on safety at sea, on access to insurance services etc. and keeps on doing so.

More importantly, a number of countries have taken on the challenge and are developing National Plans of Action to implement the SSF Guidelines. Tanzania, Namibia and Senegal already have approved such plans, and similar ones are under development in Madagascar, Malawi and will soon start in Uganda.

2022 as International Year of Artisanal Fisheries and Aquaculture provides an important platform to launch a new area of support to small-scale fisheries.

Mme Franz Nicole

Development Economist

Food and Agriculture Organisation of the United Nations (FAO)

Rome



Keynote Address (Summary)

Dr. Meryl J. Williams

Women in Small-Scale Fisheries: Proposals for a Better Future

Dr. Meryl Williams,¹ Dr. Victoria Syddall,²
Dr. Arlene Nietes Satapornvanit,³ and
Dr. Nikita Gopal⁴



¹Gender in Aquaculture and Fisheries Section, Asian Fisheries Society

²The University of Auckland, Institute of Marine Science, New Zealand

³Gender in Aquaculture and Fisheries Section, Asian Fisheries Society

⁴Gender in Aquaculture and Fisheries Section, Asian Fisheries Society & ICAR-
Central Institute of Fisheries Technology, Kochi, Kerala, India

The post-World War II fisheries boom was accompanied by transformations in human development, the economy and a backlash due to the environmental impacts. The boom had a significant but poorly documented impact on women, especially those in small-scale fisheries. In small-scale fisheries, women activists on the frontline of small-scale fisheries action felt their task was a “struggle within a struggle”¹. The post-War fisheries expansion was promoted by states and international agencies. Private, public and joint investments accelerated the dispersion of technologies, especially through industrial fishing, processing, and distribution networks. In the ensuing boom, most support initially flowed to industrial fishing development, until the importance of small-scale fisheries for food and livelihoods became increasingly evident. Even in small-scale fisheries, development assistance agencies and states’ strategies favored technology as the way to increase production, along with orienting the products to markets and supporting new governance approaches. The social domain of fisheries development was largely assumed to be fishermen working in the production nodes of the value chains. Little attention was given to human development needs more broadly, and women’s needs were overlooked except for a few technology projects in fish processing. The political economy that enabled the boom thus set fisheries on a pathway that affected the

1 Nayak, N., and C. Quist. 2016. *ICSF’s Journey with Women in Fisheries*.



outcomes for women, ranging from the way the socio-technical systems have developed to how institutions and interest groups frame priorities. Development activities aimed at helping women tended to overlook their actual and diverse circumstances and needs, and some even overburdened women with more work. A recent common response to concerns over gender inequality has been to publicly laud individual women's achievements as signs of progress. Individual women's successes, however, are not signs of transformation to gender equality in fisheries, but they most often result from the progress in women's education and, in some cases, particular social advantages. The fisheries institutions and organizations women enter remain little altered.

Overall, women in fisheries face challenges from being technologically excluded, economically ostracized and politically powerless. Fortunately, gender studies have made major methodological strides in recent decades and have much to offer to fisheries to address these three challenges. A substantial body of coherent knowledge on gender and fisheries is now available.

Based on this knowledge, we outline solutions for reversing these three conditions by proposals for including women in the new socio-technical systems, incorporating them fairly in the economy and empowering them to become politically strong. First, technological inclusion requires explicitly exploring the gender dimensions in fisheries technology dispersion and in transitions in socio-technical systems. Taking an intersectional approach, women's work in fisheries must be made visible. The gender divisions of labor and the bias towards focusing on the harvesting nodes of value chains must be redressed. Second, women fish workers and entrepreneurs are served by a large and growing body of evidence and ideas for better incorporating them into the economy. We will illustrate these, again, with an intersectional approach, with ideas from the new Cooperative Action Plan on “Decent Work and Thriving Businesses for Women in Fisheries.” Third, to gather political power, women in small-scale fisheries will need to greatly accelerate their collective action as part of the overall agenda, and as part of an overall women's fisheries agenda.

In the commoditized fisheries systems now operating, ignoring a major share of the workers and potential business people – the women in all nodes of the value chains – is no longer an option.

Dr. Meryl J. Williams

Gender in Aquaculture and Fisheries Section, Asian Fisheries Society



Keynote Address (Summary)

Emeritus Professor Svein Jentoft

Governance and the Management of Small-Scale Fisheries

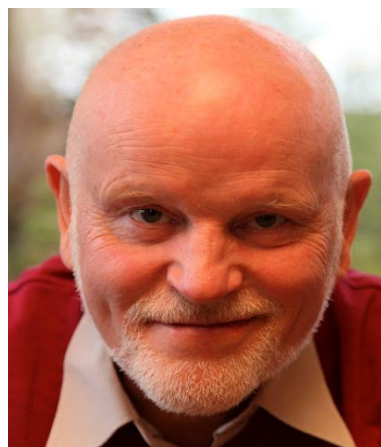
In his presentation, Jentoft discusses what is commonly understood by governance, what makes it different from management, and how these concepts apply to small-scale fisheries. He will draw on his experience of being a member of the Norwegian delegation during the Technical Consultations on the SSF Guidelines in 2013 and –

14, where the meaning and distinction between the two concepts triggered controversy for lack of clarity.

He argues that governance is broader than management, and beyond instrumentality. He also posits that although the government has a major role to play, it is not the only relevant actor in the governance system. Small-scale fisheries require an interactive, multi-stakeholder approach, which involves local people, their communities, and representative organizations. Their governance involves an open and inclusive communicative process in which the ethical principles guiding the design and functioning of institutions, and how they transpire into the daily life in small-scale fisheries communities, are deliberated. This, he argues, is what the governance lens invite to, more so than management.

Jentoft will focus on language; how we talk about small-scale fisheries and the words we use. He posits that the institutions, which we design for securing those sound governance principles like those guiding the SSF Guidelines that have an effect, are fundamentally language dependent. The SSF Guidelines are words, forming sentences and articles. Law is language. We argue and act in a governance context by means of language. Concepts highlight issues, create awareness and set the stage for what to do. Policy reforms, like those the SSF Guidelines advocate, cannot happen without language. Therefore, we need to be conscious of how which concepts to use and what they mean, including how governance is defined in small-scale fisheries situations because they matter.

The SSF Guidelines are not a technical manual, he observes. First and foremost, they invite a broad conversation about small-scale fisheries with a different vocabulary than what typically dominates a management discourse. Many of the qualities of a well-functioning small-scale fisheries governance system require deeper reflection. To eradicate poverty, there must be a consensus on what poverty is. Same with equity, justice, and human rights, neither of which is easy to define. To empower small-scale fisheries in their governance, we must be clear about what power is and how it works.





However, these issues cannot just be talked about in the abstract. They must also be defined in the concrete contexts where the SSF Guidelines are meant to function. Notably, this conversation does not precede governance but is an integral part of it. It is easy to agree on principle as delegates did during the Technical Consultation on the SSF Guidelines. For instance, the delegates did not have any reservations about human rights as a guiding principle as they did with the governance concept, even if it can be argued that human rights require governance more than management. How the principles should apply in practice, proved more of a difficult thing to agree on. Still, the SSF Guidelines luckily survived.

Emeritus Professor Svein Jentoft
Arctic University
Norway



Message from the Conference Chair

Professor M.G. Kularathne

The United Nations General Assembly has declared 2022 the International Year of Artisanal Fisheries and Aquaculture (IYAFA 2022). As small-scale fisheries provide about 40% of the world's fish production and on the global scale, small-scale fishers and fish workers account for about 90% of the people who are engaged in capture fisheries value chains, Food and Agriculture Organization (FAO) has designated the theme 'Small in Scale, Big in Value' for IYAFA 2022. According to FAO, the objective of celebrating IYAFA 2022 is to draw the world's attention to the role that small-scale fishers, fish farmers and fish workers who play a significant role in food security and nutrition, poverty eradication and sustainable natural resource use – thereby increasing global understanding and action to support them.



Sri Lanka Forum for Small Scale Fisheries (SLFSSF) has also taken steps to join hands with the global efforts to celebrate IYAFA 2022 and organized 'Fishing for Life 2022', the South and South-East Asian Conference on Small Scale Fisheries and Aquaculture. This conference aims at spotlighting the important contribution made by the small-scale fisheries (SSF) in providing society with food, nutrition and employment and, to underline the need to effectively manage this sub-sector to achieve sustainable use of resources and wellbeing of fishing communities.

SLFSSF, although relatively a young professional organization, well recognized that research on SSF is pre-dominantly scientific, with little focus on small-scale fishing communities who are confronted with an array of issues, such as lack of rights, their non-inclusiveness in fisheries decision making, lack of voice, resource degradation and declining livelihoods and wellbeing, etc. Small-scale fisheries offer lifestyles that deserve special attention of policy makers in the fisheries sector.

Fishing for Life 2022 will be an open dialogue between scientists and fish workers in South and South East Asia. This would facilitate to understand the status of resources exploited by small scale fishers and means of dealing with resource degradation and enhancing fish stocks due to various factors such as technological change, government policy and international trade. This dialogue will also highlight



and seek means of resolving diverse issues confronted by small-scale fishers, such as violation of tenure rights of small-scale fishers, poverty, non-inclusiveness, powerlessness, lack of alternative livelihood opportunities, etc.

The organizing committee of Fishing for Life 2022 conference received tremendous encouragement by local, regional and international eminent scientists and partners. Especially, as the Chair of the Organizing Committee, I am incredibly grateful to Civil Society Organization for their willingness to join the discourse without any hesitation. I wish every success that would undoubtedly ensure the benefits of hitherto intentionally or unintentionally neglected groups of the society, small-scale fishers.

Professor Mohottala G. Kularatne
University of Kelaniya



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Sustainable Use of Resources



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Prioritizing Fished Stocks for Monitoring and Management: Insights from Hilsa (*Tenualosa ilisha*) Gillnet Fishery of Bangladesh

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Abstract

The artisanal gillnet fishers of Bangladesh have been targeting the country's iconic flagship species, Hilsa (*Tenualosa ilisha*), to sustain their livelihood. Although it is evident that higher fishing pressure resulted in stock depletion of both targeted and non-targeted species, inadequate biological and catch information on fish stocks in Bangladesh limits the quantitative risk assessment for captured species. In this study, the Productivity Susceptibility Analysis (PSA) approach—a semi-quantitative risk evaluation approach that used both the intrinsic biological characteristic of stock and their susceptibility due to fishing to generate a risk score—was used to categorize the 75 species of Hilsa gillnet fishery of Bangladesh for prioritizing their monitoring and management. The results revealed that seventeen non-targeted bycatch species of Hilsa gillnet fishing from inland (7 species; e.g., *Pangasius pangasius*, *Plotosus canius*, *Sperata aor* etc.) and marine habitat (10 species; e.g., *Leptomelanosoma indicum*, *Netuma thalassina*, *Scomberoides commersonnianus* etc.) were determined to be at high risk due to fishing. Since there is a strong association between the exploitation rate (>0.5 = overfishing status) and the risk score (>1.8 = overfishing status) derived from PSA, this study suggested that species with a risk score over 1.8 needed urgent conservation actions for their long-term sustainability.

Keywords: Vulnerability, Bycatch stocks, Multispecies fishery, Conservation action

Policy Recommendations:

- The mesh size regulation of gillnets, avoidance of the usage of the bottom set gillnets, and minimum legal length regulation is highly recommended to reduce the susceptibility of both Hilsa and its bycatch stocks to fishing and thus reduce the number of overfished stocks or stocks undergoing overfishing problems.



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Applied DNA barcodes for ichthyoplankton monitoring program in Vietnam: Potential for fisheries management

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Abstract

DNA barcoding has proven a useful tool in marine species identification, especially, connecting the adults to early developmental stages such as planktonic larvae. Ichthyoplankton information plays an important role in fisheries management and conservation, especially in biodiversity hotspots, such as Vietnamese marine waters, where over 2,000 species occur. This study collects Ichthyoplankton samples in Vung Tau waters, Southeast, Vietnam in 2020-2021. Egg and larval specimens were barcoded using molecular markers (16S Rna and COI mtDNA). In a total of over 50 specimens, 13 barcoded were successfully detected, and 15 species weren't fully identified. Among them that, only 2 species of eggs and larvae matched the adult fish barcodes. The early developmental stages (eggs and larvae) of the gold stripe sardinella (*Sardinella homarus*) and fork-tailed threadfin bream (*Nemipterus furcosus*) were described. Sardine's eggs were found in August, while larvae occurred in March. For *N. furcosus*, both eggs and larvae were recorded in April. These findings can support determining the spawning seasons and sites of marine fish species, as the basis for the management and conservation of marine resources.

Keywords: Ichthyoplankton, Barcodes, Fisheries management, Vietnam

Policy Recommendations:

- These findings can support determining the spawning seasons and sites of marine fish species, as the basis for the management and conservation of marine resources.





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Assessing the length-based spawning potential ratio (LBSPR) of Scalloped spiny lobster (*P. homarus*) in Hambanthota district, Sri Lanka for sustainable fishery management

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Abstract

Small-scale fisheries largely support food security in coastal communities as a source of income for livelihoods and their well-being. Socio-economic dimensions show that the Scalloped spiny lobster (*P. homarus*) fishery in the Hambantota district of Sri Lanka has become very popular but resources have declined due to various factors. Therefore, the current assessment provides information on spawning stocks and fisheries to inform existing management regulations. Length frequency data collected from major lobster landing sites in Hambantota district during 2007 to 2019 and Life History Data from published literature were used to estimate LBSPR. As lower (LRP) and upper reference points (URP); 0.3 and 0.4 are considered. The calculated LBSPR for the years 2007, 2017, 2018, and 2019 were 0.22, 0.2, 0.27 and 0.21 respectively. findings revealed that LBSPR was lower than LRP during the study period. The LBSPR during 2014 -2015; were 0.18, 0.17, respectively and fell below the threshold level of 0.2 (20%). Relative fishing pressure (F/M) >1 (1.39-2.76) for the respective period. The estimated LBSPR and F/M values explain the overexploitation status of the stock. Strict enforcement of regulations with a larger minimum legal size (70 mm) is recommended to improve the spawning stock required for the sustainability of the fishery.

Keywords: Scalloped spiny lobster, *P. homarus*, LBSPR, Relative fishing pressure

Policy Recommendations:

- Strict enforcement of the existing regulation.
- Amend the minimum legal size (from 60 mm to 70mm in carapace length).
- Stop issuance of new licences except for the current fishes.
- Effort control through extending closed season.





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How do smallholder fishers get by with climate change impacts in the Galle district, Sri Lanka

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Abstract

Hundreds of millions depend on small-scale commercial and subsistence fishing for income, livelihoods, and food security. Climate change poses considerable obstacles for small-scale fisheries and is expected to influence the lives of those who rely on it significantly. The main objective of this investigation was to understand the small-scale fishers' perceptions and awareness of the effects of climate change in the Galle district. A structured questionnaire was used to collect data from 270 respondents. The investigation was carried out utilizing a quantitative research methodology, and the gathered information was examined using descriptive statistics and binary logistic regression. Likewise, the study was mainly based on both primary and secondary data. The result revealed that 76.3% of respondents were aware of climate change, while 23.3% were not. The awareness of climate change was used as the dependent variable. Age, employment status, primary occupation, fishing experience, and membership in a fishing association were among those independent variables that were statistically significant. Still, education, annual fishing income, and marital status were not. As a result, the final finding was that age, primary occupation, fishing experience, and membership in fishing associations positively influenced the dependent variable.

Keywords: Climate change, small-scale fishers

Policy recommendations:

- Communication clearly with stakeholders, including the public, on how climate change will affect fisheries.
- Protect ecosystem
- End environmentally harmful subsidies
- Extend the use of rights-based management system
- Prohibition of fishing by destructive methods. They are imposing fines on fishermen who do so.





Impact of Policy, Technology, Trade, and Development on SSF



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**Research and simulation of ITS (intelligent transportation system)
Enabled tracking and control system for Small Scale Fishing boats**

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Abstract

Guided navigation will prevent SSF boats from rescue situations such as distress conditions. There is no proper system to overcome this issue in the Sri Lankan fishing industry SSF Boats. Hence, the main objective is to implement an ITS-enabled navigation system with real-time communication to reach boats to a safer destination. Secondary data with a literature review were used as the main research methodology for this implementation. This proposed system support is “guided-self-driven” till the boat back to the shore in a fishing session. The “Control Center” that can guide, Manage and communicate with the fishing boats, using boat-units (made up of a Raspberry-Pi-4 circuit board installed Raspberry-Pi-Lite-OS) are fixed to the boats which are embedded in a GPS unit and a terminal. The use of GIS technology and visualization maps will guide to reach the pre-decided destination. The boat unit is connected with the “Intraoperative-layer” for the positioning, communication, and identification of the boats through GPS satellites and Internet connection (Inmarsat Satellite / GSM). “Bayesian-process” ($W_t = AW_{t-1} + Y_t$) is used to detect and trace boat units over real-time motions. The Control layer operates over an ITS-system, which functions over the establishment of the links between various units of the system and their functions to take appropriate decisions for the boats during the fishing session.

Keywords: Intelligent Transportation system (ITS), Maritime Transportation, Fishing Boats Tracking and Control system, Global Positioning System

Policy Recommendations:

- Provide guided navigation to SSF boats during the fishing session and distress situations for fisheries and relevant stakeholders.



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Localization (positioning) system for small scale fishery boats in Sri Lanka

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Abstract

Showers or thundershowers and winds will occur at sea areas in Sri Lanka. Hence, small scale fishery boats are facing significant distress situations on the sea and estimating the boat position accurately will enable one to manage the distress of boats from the central management system. Determining the position of a target boat is one of the most important technical challenges in the navigation system. The underwater wireless sensor network will play a massive role in it and it is needed to use an accurate positing (localization) method to estimate the position of the boat. Secondary data were used for evaluating trends of localization method using a Literature review (existing body of work) and Virtual Instrumentation has been used for realizing the accurate position of the boat. When underwater challenges are considered, the TDOA localization technique is the most suitable technique for underwater localization. Thus, it has evaluated the Iterative (Taylor) and Closed-form method (Chan method/TSWLS) for the Time Difference of Arrival. The CRLB has been used to take the best result of the unbiased estimator to find the target position (boat). Hence, this proposed system, as an impact of the technology, will help to the development of the SSF industry as the impact of the technology.

Keywords: Localization, Underwater Wireless Sensor Networks (UWSNs), Time Difference of Arrival (TDOA)

Policy Recommendations:

- Determine the location of small-scale fishery boats while in distress situation for fisheries and relevant authorities.



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**Fish Processing Technologies and Fish Consumption Patterns among
Inland Reservoir-based Fishing Community in Kalawewa Reservoir, Sri
Lanka**

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Abstract

A self-administered questionnaire was used to study fish processing technologies and fish consumption pattern of the fishing community in Kalawewa reservoir by randomly selecting 50 fishermen from 3-fish landing sites. Binary logistic and ordinal logistic regression were conducted for statistical analysis. There were 26 that fishermen engaged in fish processing, while 38% preserved fish at monthly frequency and all practiced fish smoking. 53% engaged in salting & sun-drying. Mostly Tilapia was used for smoking for which household smoking kilns were used, which generated an extra income for 46% fishermen. The use of fish processing technologies had a significant positive relationship with the availability of fish storage facilities ($p < 0.05$). Inland fish consumption/week and number of fishing days had a positive relationship. 42% consumed 2-3 kg of inland fish/week, whereas 66% rarely consumed marine fish. Cooking of fish with chili & other spices (Mirisata) was practiced by all, followed by the use of spices & coconut milk (76%) and cooking after deep frying (76%). Preparation of devilled-fish (46%) and sour-fish (36%) was less common. The major challenge was low fish harvest. Fish processing was positively related to fish availability, while fish consumption was to the number of fishing dates. Smoking of fish was the main processing technology practiced.

Keywords: Cooking methods, drying, smoking, Tilapia

Policy Recommendations:

- It is essential to provide fish storage facilities for surplus catch especially in aquatic resources, where a surplus catch is often encountered.

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Factors Affect the Consumption of Marine Fish among Households in Galle District

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Abstract

With the rapid change in people's economic situation and lifestyle in the present scenario, their consumption patterns are also changing. With the current economic downturn, marine fish prices are rising sharply. When consumers purchase marine fish for consumption, various reasons influence their buying preferences and willingness to pay. This study intended to explore the potential factors affecting consumers' choice to consume marine fish among households in the Galle district, Sri Lanka. The study was carried out in different markets in the Galle district. Data related to socio-economic factors of households, marketing factors, and product characteristics were collected from 200 respondents in 5 markets across 5 Divisional Secretariat Divisions in the district. The quantity of marine fish purchased per month was significantly affected by the timeline of buying fish ($p < 0.01$), the number of family members ($p < 0.01$), monthly household income ($p < 0.01$), expense for marine fish per month ($p < 0.01$) and promotional tools ($p < 0.01$). The other factors which are age, distance from home to the nearest fish market, method of buying fish, fish market near to home and type or nutrition of the fish do not significantly affect to the quantity of marine fish purchased per month.

Keywords: Consumption, marine fish, households, households' income

Policy Recommendations:

- Charge reasonable prices by fish-mongers who come to doorstep for selling marine fish.
- Increase the demand for marine fish through the adoption of marketing strategies.
- Providing adequate marine fish supply not only in coastal areas but also in all other areas in Galle district.

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Aquitourism: Synergy between aquaculture and tourism

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Abstract

Aquitourism, aquaculture-related tourism, a new concept to explore, is an untouched concept practiced in many parts of the world as an alternative tourism activity. Our approach was to investigate aquitourism sub-sector and identify its value chain. The case study approach was instrumental to identify and explore the aquitourism ventures practiced in different agro-ecological zones of Sri Lanka. Research locations were south (Balapitiya, Koggala, Dodanduwa, Rekawa) and the west coast (Negambo and Kalpitiya) of Sri Lanka. Field observations and participatory methods were key to identify the nature and structure of the aquitourism value chain. Aquitourism value chain is short, complex and downstream is linked with leisure tourism industry. Upstream comprised of small-scale fishermen and hoteliers. Service providers bridge upstream and downstream of the aquitourism value chain. Sport and recreational fisheries and guided tours are taking place within an area used for aquaculture. Upstream end enjoyed the returns for the fish catch as well as services provided. Tourism service providers empower local communities through employment creation and multi-functional utilization of natural resources. Resource use conflicts were common, and community-based participatory approaches were often applied to manage the issues and problems. Aquitourism as a multi-use concept is to be developed to empower the rural aquaculture sector and improve revenue for its stakeholders.

Keywords: Aquitourism, Rural Aquaculture, Value Chain

Policy Recommendations:

- Identify and empower aquitourism value chain.
- Educate aquitourism value chain actors on sustainable tourism management.
- Develop community-based participatory resource management mechanism.

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An architectural analysis of fish and seafood markets of the South of Sri Lanka

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Abstract

Fish and seafood, central to the Sri Lankan diet, are caught locally and sourced freshly from diverse local market places which are usually the least attractive places for shoppers. Our aim was to perform an architectural analysis on fish and seafood markets and specific objectives were identification of sustainable status of the fish and seafood markets, and to explore a sustainable design for fish and seafood markets in which socio-cultural wellbeing is ensured through the creation of market space. The case study approach was instrumental in data collection and three main market types were purposely selected for the study. The case studies were Public market of Galle, Dondra port fish market, Dodanduwa and Ambalangoda fish markets. Participatory observations, site visits, in-depth interviews with market players and storytelling exercise with managers and administrators were key data collection tools. Location, building design and structure, neighbourhood environments, market infrastructure, materials used, and people engaged were explored and compared among the case studies. The Fish markets continue its history as a place for the people, an open and vibrant environment where the daily fish trade and other marked trades can expand freely. Sustainable status and green rating were ranked as poor for all except Galle public market, a historic Dutch building. Market infrastructure and materials used were mismatched with the purpose and significantly affected performance.

Keywords: Architectural design, fish markets, green buildings

Policy Recommendations:

- Architectural designs and green building concept are an essential part in design fish markets.
- Market structure essentially needs to include fish by products and waste management mechanism.
- Market design in line with the Sri Lankan green rating system.

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Consumer preference towards value added canned fish products: A case of Bentota divisional secretariat in Galle district

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Abstract

The market competition for the food business requires the development of differentiated value-added products to meet the demand of consumers. To increase the production and marketing of value-added canned fish products, the study offers several ideas for canned fish production and effective canned fish marketing based on consumer preferences. Therefore, this research aims to find out factors affecting consumer preference towards values added canned fish products and identify the most effective marketing activities. This information will provide insight into the management of the small-scale fisheries sector by capturing more local and international market share and enhancing small-scale sector wellbeing. A total of 150 canned fish product consumers were acquired in Bentota divisional secretariat through the convenient sampling technique while adapting the online questionnaire survey method to perform the multiple regression analysis and descriptive statistics. Results of the multiple regression analysis indicated that socio-economic factors such as family size and average monthly income, Product characteristics such as quality, price, taste, and freshness significantly impact consumer preference. Descriptive statistics were used to identify the most effective marketing activities. The results revealed that discounts/offers are the most effective marketing activities and social media, television advertising and word-of-mouth are other effective marketing activities respectively

Keywords: consumer preference, development of small-scale fisheries sector, multiple regression, value-added fish products

Policy Recommendations:

- To promote the small-scale canned fisheries sector, it is important to consider consumer preferences while focusing on effective marketing attributes in order to capture both the domestic and foreign markets.



- With better guidance and management, the small-scale fishing sector in Sri Lanka can play a vital role in food security and poverty eradication.
- Securing and increasing the contribution of small-scale canned fish producers to the fisheries sector, may contribute to the sustainable social and economic development of the small-scale fisheries sector in Sri Lanka.
- This study provides information on consumer preferences and the most effective marketing strategies for the development of the small-scale fisheries sector in Sri Lanka.



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Status and Challenges of Provenance in Sri Lankan Small Scale Tuna Industry

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Abstract

This study intended to explore the present status of the provenance of the export-oriented tuna value chain, identify the challenges to the provenance of the small-scale tuna fish industry and identify the potentials for a decentralized provenance mechanism. A mixed method approach was utilized by gathering quantitative data from Ceylon tuna exporting companies and qualitative data were gathered through focus group discussions with downstream, midstream, and upstream actors. Major gap in the existing system is that there is no real-time data uploading and authentication by upstream and midstream actors. The exporter dominates the traceability and authentication process lowering the credibility and increasing chances of IUU fishing, financial frauds, mislabeling, increased food miles, lower worker welfare, and quality and safety verification issues. Challenges such as lower technological literacy, unaffordability of technological instruments, unawareness of the importance of responsible fisheries, low competency and the income orientation of small-scale fishermen limited the decentralized provenance mechanism. Potentials for decentralized provenance included the demand for traceable, authenticated, and ethically sourced products from high-end markets, sustainability certification, and Fisheries Improvement Projects. These certifications will make a decentralized authenticated provenance mechanism a mandatory requirement.

Keywords: Provenance, small-scale fisheries, tuna, value chain

Policy Recommendations:

- It could be recommended to improve fishermen's knowledge about responsible fishing, traceability, and transparency through community-based fisheries organizations.
- Facilitate technological equipment and literacy on handling traceable equipment at the sea.
- Strengthen and monitor the existing policies to demotivate IUU to maintain healthy fisheries stock at the sea.



- Establish a national small-scale fisheries action plan for community empowerment and to strengthen the post-harvest value chain.



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Fair & sustainable fish and seafood value chain: A case of Sri Lanka

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Abstract

The study aimed to investigate the fish and seafood value chain of Sri Lanka. Key concerns were VC architecture, governance and income distribution. Special attention has been paid to human rights, gender involvement, climate change, and the COVID-19 pandemic, at different points in the value chain. Data collection and value chain analysis enriched with a participatory visual mapping approach along with social network research, visual research methods and participatory statistics in eight fisheries districts. Short and fragile supply chains cater to the food security needs of primary producers and their communities while supply chains were stretched to provide fish and seafood to more distinct communities. Value addition and creation convert supply chains into complex, lengthier value chains. Regional value chains are short; they contain few nodes or a limited number of players. Few or no middlemen or middle players allow for fair income distribution along the value chain where ends, primary producers, and consumers receive appropriate payments and rates. Smooth information flow through many players allows for transparent decision-making, especially price decisions. In contrast, lengthy value chains are complex, close to network architecture, and contain many players upstream weakly connect with downstream, complex nature disturbs the smooth flow of information and unfair income distribution

Keywords: Fair, sustainable, Seafood, value chain

Policy Recommendations:

- Short seafood value chains operate locally and provide fair income distribution compared to lengthier regional and international value chains. Strengthening value chain networks through information channels are essential for equitable distribution of income and removing bottlenecks.

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The Xpress Pearl Shipwreck Impact on Dried Fish Consumption in Sri Lanka

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Abstract

The Xpress Pearl Shipwreck occurred in May 2021 near Colombo port, Sri Lanka affected marine fish and dried fish consumption due to the release of toxic compounds into the sea. This study was focused to identify the impact of the shipwreck on dried fish consumption in Sri Lanka. A questionnaire was distributed online to marine dried fish consumers on the whole island using convenient sampling technique and resulted in a sample of 125. The results show that the average reduction of the number of consumption days, amount and the expenditure on dried fish per week were 22%, 7% and 20% respectively. However a significant difference could be observed only in the average number of consumption days and the average expenditure per week (Paired t test; $p=0.002$, $p=0.045$ respectively at 5% significant level). Majority of the consumers have stated that they believe that dried fish were processed using fish containing the toxics released by the ship and hazardous to consume while 20 consumers have ranked dried fish in first rank as the alternative for fish after the shipwreck. However, 62% of the consumers have refused consuming dried fish assuming that dried fish processed from toxic fish would come to the market in the near future.

Keywords: Xpress Pearl, Dried fish, consumption

Policy Recommendations:

- Food safety is one of the key components considered by the consumers. Hence policies which ensure food safety at all the circumstances is a timely need.

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Fishery Value Chain Analysis in the Southern Coast of Sri Lanka

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Abstract

Value chain analysis can be effectively utilized as an important tool for identifying the required modification for upgrading the fishery industry. This study aims to identify key economic agents, their roles and functions in fisheries value chains, and estimate monetary flows through various marketing channels. The data was collected by using in-depth interviews. A minimum of five agents were sampled in each node of the tuna and reef fishery value chains giving a total of 134 individuals and five exporting companies. Fishers and exporting companies were selected using stratified purposive sampling, and other stakeholders were selected by snowball sampling. The exporting channel consisted of fishers, wholesalers, company suppliers and companies. The domestic channel is aligned across fishers, wholesalers, retailers and consumers. Domestic channels are primarily harvesting driven and export channels are market-driven. The fishers and exporting companies incur the highest marketing cost in the domestic channels and export channels, respectively. The proximity to the landing site is an important factor in determining consumer prices. The highest fisher to retailer price spread was reported in export channels. Due to the wider price asymmetries, the deflated prices for producers and inflated prices for consumers can be observed. Even though most of the market channels are profitable, the distributions of gains or losses along the channels are not proportional and fair. Therefore, it is crucial to introduce a proper pricing mechanism and market regulations. For instance, fair distribution of retail prices should be insured within the value chain system.

Keywords: Cost, Fish, Marketing, Price, Value chain

Policy Recommendations:

- Implementing proper market pricing mechanisms or enough price regulation

Retailers' perception towards sanitation and sustainability of fish packing materials: A study in Galle district

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Abstract

Fish packing materials play a fundamental role in assuring safe delivery of fresh fish to the end consumer in good condition. The objective of this study was to evaluate retailers' perceptions of the sanitation and sustainability of fish packing materials in Sri Lanka. The study was carried out in collaboration with the major fishing harbours of Galle district in Sri Lanka; Galle, Hikkaduwa and Ambalangoda. Seventy-five fish retailers who regularly visit these 3 harbours participated in this study; 25 fish retailers were randomly selected from each harbour. Fish retailers had an awareness of the sanitation of fish packing materials with a perception of sustainability ($P < 0.05$). The majority of retailers (76%) were willing to use standard, eco-friendly packing materials for packing or storing fish. A positive relationship existed between sanitary practices and sanitation awareness ($P < 0.05$). The perception of sustainability wasn't statistically associated with socio-economic factors such as age, gender, education and income level or marital status ($P > 0.05$) of retailers. According to the results, retailers were aware of the sanitation and sanitary practices of fish packing materials. Even if retailers were unfamiliar with the theoretical concept and had a basic idea of how packing materials contribute to a country's sustainability.

Keywords: Fish Packing Materials, Fish Retailers, Sanitation, Sustainability

Policy Recommendations:

- Sustainability, Fish packing materials, awareness programs of fish retailers.

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Factors Affecting Purchasing Decisions of Fresh Fish in the Samanalawewa Area

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Abstract

Sri Lanka is a place with enough fresh fish resources for consumption. Rathnapura district is one the highly affected by malnutrition and inequality issues., which paves the avenue to be based this research in the Rathnapura district. To examine this situation, data was collected from the population of Imbulpe and Balangoda DS division, and a sample of 230 consumers including eight Grama Niladari divisions used a structured questionnaire. Data were analysed using a Choice Experiment approach to determine respondents' best choices among the three attributes. The three attributes tested in this model are the type of fish species, willingness to pay, and the status of fish as fresh or frozen. The preliminary result of the research indicates that the model results are highly significant. All three attributes used in the model are significant at all levels except only one level of the fish type attribute. Omnibus Tests of model coefficients show the overall coefficient of chi-square was significant at the $\alpha = 0.01$ level. The most important attributes reflect higher positive B values and "Freshness", Price is "Rs. 1200", and "Korali Fish" was the most important attribute at all. Results also indicate that people of this area purchase fewer fresh fish to fulfil their daily nutrition requirements and that residents can be categorized as low-income groups.

Keywords: Fresh fish consumption, Nutrition, Choice experiment, Rathnapura District

Policy Recommendations:

- Providing adequate infrastructures to fishers for expanding production and making available at an affordable price for poor people.
- Propagate fish species in small reservoirs to enhance the fresh fish experience in rural.
- Promoting fish into the diet is a strategy to control protein energy malnutrition and micronutrient deficiencies in rural areas.



- Raising awareness by connecting different roles in the society such as pregnant mothers, midwives, school teachers, and lactating mothers.



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Risk Assessment in the ornamental fish farmers in Colombo district in Sri Lanka: Especially in the context of COVID-19 pandemic

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Abstract

Systematic risk management is critical in dealing with today's corporate difficulties. With the COVID-19 pandemic, ornamental fish farmers' have faced different types of risks. In this study, the types of risks associated are grouped into six categories, namely macro level risk, demand risk, supply risk, quality risk, risk on assets and infrastructure, and financial risk, to find out the significant risks and socio-economic factors of ornamental fish farmers. Moreover, it analyses the association between production category and risk assessment of ornamental fish farmers. As a result of the study, the quality risk is at a considerably high level, and a risk on assets and infrastructure is also significantly affected during the COVID-19 period. Concerning socio-economic factors, male farmers were mainly participating in ornamental fish production, and also a higher number of farmers had education levels up to GCE Advance Level. The farmers had more than 10 years of experience in the field. Furthermore, the production capacity of most of the farms belonged to group IX (8333-83333 fish per month). Moreover, only macro-level risk showed an association with the production capacity. While the farmers with a production capacity of 833-8333 fish per month (V group) dominantly faced the macro-level risk, other risks didn't show the association with the production capacity.

Keywords: Colombo district, COVID-19 pandemic, Ornamental fish farmers, Risk assessment

Policy Recommendations:

- For overcoming the quality risk, farmers should be graded based on their fish quality and quantity to prevent the refusal of their products and could be introduced the stable price for the fish based on this grading system.
- Hence the government should be facilitated to establish the fish quality checking team in Sri Lanka.
- the government should develop the proper system to access the particular markets for selling their products and conducting motivational programs to contribute to increase the involvement of ornamental fish production, marketing and exporting.



Poverty, Rights, and Rural Aquaculture



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Significance of dried fish value chains in supporting the wellbeing of fishing communities: Insights from Sri Lanka

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Abstract

Dried fish is an important sub-sector within small-scale fisheries that support the livelihoods and wellbeing of fishing communities in Sri Lanka. Dried fish production, trading, and distribution activities are structured as value chains. However, the significance of these value chains in supporting the wellbeing of fishing communities is rarely acknowledged and poorly understood. Toward addressing this knowledge gap, this presentation aims to draw from value chains and social wellbeing literature to examine how dried fish value chains underpin the wellbeing of fishing communities in Sri Lanka. The study employed qualitative case study approach to develop two place-based dried fish value chain case studies based on Kalutara and Kantale areas. Study findings revealed the scale, structural organization, and gendered roles within the study value chains. Furthermore, the findings highlighted the nuances of how these value chains support the material, relational, and subjective dimensions of wellbeing of women and men participating in them. Insights emerged through this study are of relevance to understand the entry points to strengthen dried fish value chains, particularly in the context of Food and Agriculture Organization's Voluntary Guidelines for Securing Sustainable Small-scale Fisheries.

Keywords: Dried fish value chains, Gender, Social wellbeing, Fishing communities

Policy Recommendations:

- Fisheries policy and management should aim to assess the scale of the dried fish sub-sector by expanding current enumeration efforts to capture all types of dried fish operations, including the seasonal home-based artisanal production, and thereby reposition dried fish value chains as an important segment of fisheries post-harvest sector in Sri Lanka.

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GnRH Induced Breeding and Larval Growth Performance of *Ompok ceylonensis* on Phytogenic Feed Additives

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Abstract

Butter catfish (*Ompok ceylonensis*) is an underutilized endangered species in Sri Lanka and has not yet been introduced to the aquaculture industry due to insufficient research data on hatchery rearing. Hence, in the present study, an attempt is made to observe GnRh induced breeding and rearing of larvae with phytogenic food additives. Brooders were successfully bred at Dambulla Fish Breeding Center using GnRh at a rate of 0.25ml/kg for males and 0.5ml/kg for females. The larvae were fed three types of formulated diets T1 (Moringa oleifera added), T2 (Alternanthera sessilis added) and T3 (control) at two-hour intervals throughout the day in replicate for three weeks. Specific growth rate, weight gain and survival rate were measured. Weight gain and specific growth rate (<0.05) were significantly increased in both feeds with added phytogenic additives. The highest survival rate was observed at T1 and the lowest at T2. In conclusion, it can be found that GnRh can be successfully used for butter catfish breeding and frequent feeding of compound feed with added Moringa and Alternanthera is advisable to improve growth as well as reduce cannibalism.

Keywords: *Ompok ceylonensis*, induced breeding, larval rearing, phytogenic feed

Policy Recommendations:

- To increase the nutritional level of people by consuming freshwater fish.
- Low cost on feed ingredients as they can easily be found.
- Easily can be popular among small-scale farmers.
- Can increase the livelihood of the people by generating job opportunities such as feed manufacturing.

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Assessing climate change vulnerability in Sri Lankan coastal fishing communities

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Abstract

The global climate change vulnerability index has ranked Sri Lanka among the highly vulnerable countries to climate change in the world. The vulnerability to climate change depends upon three key elements: the frequency and magnitude of exposure to external shocks, the degree of sensitivity to those impacts, and the adaptive capacity of society. The survey was administered to assess the intensity of vulnerability to climate change and its impacts on the coastal small-scale fishing communities of Sri Lanka. A stratified random sample of 200 fishers residing in 8 coastal districts was drawn and interviewed in 2021. The vulnerability index relevant to each district was developed. The results demonstrated that the vulnerability indexes differed and have been perpetuated due to heterogeneity in physical and financial resources within the fishing communities in coastal districts of Sri Lanka. Distinctly found that fishers in the Jaffna peninsula and adjacent islands were more vulnerable and less adaptive mainly due to the inability in operating mechanized boats and advanced fishing gear technologies while Hambanthota and Negombodistricts fishers were low vulnerable due to practiced coping strategies, high-level of resource availability, diversified fishing technologies, and availability of alternative livelihoods.

Keywords: Adaptation framework, Climate change, Fishing communities, vulnerability index

Policy Recommendations:

- It is essential to establish a collaborative network among government private and community institutes to predict the negative consequences of climate change and immediately mitigate the negative impact of climate change.



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Growth performance and survivability of *Catla catla* (Cyprinidae) fry fed with different levels of two cultivable mushrooms in Sri Lanka

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Abstract

Catla catla considered as one of the predominant major carps cultured in Sri Lanka. However, the survivability and growth performance of *C. catla* have been challenged for many reasons in both large-scale and small-scale fisheries. Therefore, this experiment was conducted to evaluate the growth performance and survivability of *C. catla* fry fed with 2% and 5% oyster mushroom (OYM) (*Pleurotus ostreatus*) and abalone mushroom (ABM) (*Pleurotus cystidiosus*) incorporated diets (Crude Protein (CP)-37%, Crude Fat (CF)-9%) commercial carp feed was used as a control feed. 3000 catla fry (Initial average weight- 0.37 ± 0.8 g) were allocated among 15 tanks and fed to 12%, 8% and 6% of their body weight for 1-4 weeks, 5-8 weeks and 9-12 weeks respectively. Total body weight (TBW) and total body length (TBL) were recorded biweekly. The final TBW (1.29-1.66 g), TBL (4.67-5.02 cm), weight gain (243.94%-340.32%) and specific growth rate (1.65-1.89) did not show any significant ($p < 0.05$) differences among the treatments. FCR of the experimental diets varied from 2.34-2.37 where the values were not significantly different ($p < 0.05$). However, the fish fed with 5% ABM had the highest ($p < 0.05$) survival rate (95.832%), whereas the control had the lowest (85.161%) survival rate. According to the results, inclusion of ABM as a feed ingredients increases the survival of *Catla catla* fry.

Keywords: Abalone mushroom, *C. catla*, oyster mushroom, survival

Policy Recommendations:

- 5% inclusion of abalone mushroom into fish feed can be increased the survival rate of *Catla catla* reared in NAQDA centers.



DOI: 10.13140/RG.2.2.29064.08962

***Hermetia illucens* larval meal as a fishmeal replacer in *Xiphophorus maculatus* diets**

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Abstract

Platy fish (*Xiphophorus maculatus*) is one of the major export-oriented ornamental fish species being cultured in Sri Lanka. Therefore, this study aimed at the development of a feed for *X. maculatus* by replacing expensive fishmeal with *Hermetia illucens* larval (HIL) meal. Four, iso-proteic (40%), iso-lipidic (6%) and iso-caloric (16.5 MJ/Kg) feeds were developed by replacing 200 g of fishmeal at 0% (control), 12.5%, 25% and 37.5%. Platy fry (total body weight-0.017±0.001g, total body length-1.016±0.017 cm) were allocated among tanks (stocking density-6 fries/ft²) and fed to apparent satiation thrice a day for 81 days. The 12.5% and 25% HIL diet groups had the highest final body weight (1.001±0.035 g and 1.022±0.034 g) and final body length (3.613±0.038 cm and 3.609±0.036 cm), whereas the control and 37.5% groups had significantly lower body weights (0.844±0.037 g, 0.583±0.017 g) and lengths (3.468±0.046 cm, 3.059±0.026 cm). Body weight gain followed a similar trend while the specific growth rate and body length gain were significantly higher in the control, 12.5%, and 25% groups. The feed conversion ratio was extremely poor (3.350±0.147) in 37.5% HIL dietary group compared to the other dietary groups. Therefore, HIL meal can be used to replace 25% or 59.6 g of fishmeal in *X. maculatus* diets for optimum growth performance.

Keywords: Black soldier fly larvae, growth performance, platy fish, replacement

Policy Recommendations:

- Replacement of fishmeal by *Hermetia illucens* larval meal is recommended in the diets of *Xiphophorus maculatus*. And more research is required to verify the effect of *Hermetia illucens* meal on the internal body parameters of fish.



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Sugargarze (*Sorghum bicolor*) seed meal as a replacer for maize (*Zea mays*) in *Catla catla* feed

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Abstract

Small-scale aquaculture in Sri Lanka is heavily affected by soaring prices of maize and its scarcity, leading to a hunt for commercially viable alternative energy sources an improved sorghum variety (*Sorghum bicolor*) Sugargraze is a fodder species having a protein content of 10.8% and energy 4007 Kcal/Kg, making it a potential candidate. This study aimed to investigate the effect of replacing maize with Sugargraze seed meal on the growth performance of *Catla catla* fry. A total of 450 fry were randomly distributed into five treatments with triplicates. Five iso-preotic (38%), iso-caloric (3100K Cal/Kg) and iso-lipidic (4.8) diets were formulated by replacing Maize meal, with Sugargraze meal (100%, 75%, 50%, 25% and 0%). The highest weight gain (0.58889) and length gains (0.38194) were recorded in 50%, followed by 25% and control diets. The inclusion levels 75% and 100% caused a significant reduction of growth in Catla larvae, which may be due to the high tannin content native to Sorghum. In conclusion, Sugargraze seed meal is an effective and economical replacement for corn-meal up to 50% in Catla larvae production systems.

Keywords: Sugargraze, *C.catla* fry , growth performance , replacement diet

Policy Recommendations:

- Sugargraze seed can be used partially replace of maize in the production of fish meal – policy recommendation.



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Dried fish making as a means of building resilience against fishing related vulnerabilities in small-scale fishing communities of Southern Sri Lanka

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Abstract

Small-scale fishing communities are increasingly being affected by various shocks, changes, stressors and uncertainties, raising questions about their ability to cope with them. On another front, fishers are entangled in a poverty trap and escape from it is difficult due to their poor access to diverse livelihood capitals. Fisher populations have been compelled to confront these diversity of shocks and adapt themselves in order to “remain in the game”, which is an arduous task, not easily attainable. Thus, fishers need to develop strategies to effectively cope with shocks and stresses, or in other words, to be resilient. Dried fish making is one of the traditional household activities of fisher women in Sri Lanka. In this context, a review was carried out to identify the capability of “dried fish industry” towards building resilience against diverse vulnerabilities confronted by small-scale fishers. Based on the reviewed literature it could be argued that dried fish would contribute towards building resilience in three ways; i) as a source of supplementary income which tend to smoothen inter-temporal flows of daily fishing incomes. ii) as a cheap source of animal protein in meeting the nutritional needs of vulnerable people, and iii) by playing a major role towards women empowerment and accumulating social capital.

Keywords: Vulnerability, Resilience, Dried Fish, Small-scale Fisheries

Policy Recommendations:

- Promote fish processing in building resilience among fishing communities.

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Setting up a breeding protocol to enhance breeding performance of potential food fish Green chromide (*Etroplus suratensis*) under captive conditions in cement cisterns

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Abstract

Green chromide (*Etroplus suratensis*) has a reputation as a healthy and tasty food fish species and a cheap source of protein for rural poor in Sri Lanka. Though once common in reservoirs and brackish waters, their populations have declined dramatically, hence, require stock enhancement, which will secure rural food security and income. Aim of the present study was to setup a breeding protocol for *E. suratensis* for a mass production system applicable for small-scale farming. Six pairs of matured brooders length ranging females 13cm to 17cm and males 18cm to 19cm were stocked into flow-through individual cement cisterns for 1 & 1/2 months, and fed with 42% CP containing commercial feed. Tank bottom and substrates were cleaned to stick eggs. All pairs spawning to larval development were observed at 28.7 0C± 0.41oC, dissolve oxygen 6.21 mg/l ± 0.36mg/l, and pH at 7.22 ±0.39. Elliptical, pale yellow colour, 2.4x1.1mm size eggs were laid 350-450 per couple and hatched after 66-72 hours. Approximately seven days post hatching, 5mm long larvae were fed with Daphnia and algae, and the breeding interval varied from 15-18 days during the experiment. In conclusion, this study provides baseline breeding conditions for green chromide under the captive conditions in Sri Lanka.

Keywords: *Etroplus suratensis*, Captive breeding, Relative fecundity, Spawning

Policy Recommendations:

- Developed breeding protocol for Green chromide (*Etroplus suratensis*) under captivity is successfully conducted and recommended for the production of offspring fish farming.



- As a healthy and tasty food fish, it is important to conduct further research on culture practices such as pond fish farming.





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Determinants of the catch rate of inland fisheries: A case from the Periyamadu Reservoir

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Abstract

Periyamadu reservoir is a medium-sized reservoir in the Mannar District. The fish production in this reservoir as in most medium-sized reservoirs in Sri Lanka is frequently associated with low catch rates. As such, the focus of this research is on how a fisherman's catch rate varies with demographics, household characteristics, fishing-related factors, the role of the fishery society in fishing, challenges such as crocodile disturbances, marketing constraints, a lack of tools, and perception of low yield. Using pre-tested questionnaires, data on 45 fishermen and 45 non-fishermen were obtained from the Periyamadu Grama Niladari Division. Heckman's two-stage model was used to discover the elements that influence fishermen's catch rates as well as their desire to engage in inland fishing. The results suggested that household size and income had a significant impact on the decision to engage in inland fishing, whereas income, obstacles such as crocodile disturbances, and the perception of low yield had a significant impact on fishermen's catch rate. In addition, training programs, fishing days per week, and marketing limits all had a substantial impact on fishermen's catch rates. As a policy recommendation, the study proposes effective training programs as a means of increasing productivity of medium-sized reservoirs.

Keywords: Inland fishing, Catch rate, Constraints to inland fishing, Heckman's two-stage model

Policy Recommendations:

- Government should create robust market structure for inland value-added products in every district of the country.
- The government ought to pay more attention to undeveloped reservoirs. It improves inland fisheries industry results. (Funding, free stocking, and subsidies for fishing equipment must be taken into account).



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Division of labor mapping in rural aquaculture value chains

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Abstract

The fisheries industry plays a major role in social and economic well-being. The objectives of this study are to identify both visible and invisible roles of men and women in each node and mapping rural aquaculture value chains on the division of labor. Dry fish value chain, Maldives fish value chain, and ornamental fish value chain were selected and data were collected by rapid market chain analysis. Focus group discussions were held in Negombo, Rathnapura, and Puttalam districts. The selected sample included fishermen, women, primary processors, wholesalers, and traders. Primary activities, secondary activities, and inbound and outbound logistics are considered. Women's involvement in fishing is considerably low but involvement in primary and secondary activities is higher. Division of labour mapping disaggregates the gender division of labour to allow an understanding role of women in value chain nodes. Value chain node activities such as post-harvest handling, biosecurity practices, and trade feminized in rural society strengthen women with financial and other skills. Since women's involvement in the fishing industry recommend the livelihood development of rural fishing areas Well-developed extension services, guidance, and gender equity are needed in the fishing industry to gain higher productivity, smart farmer hubs and the application of new technology which are also effective pathways.

Keywords: Aquaculture, biosecurity, Food security, labour mapping

Policy Recommendations:

- This study identifies the dedication of the fishermen in the Sri Lankan fishing industry and their disappointment due to the lack of resources and the relevant skills to be successful in the fishing industry.
- Through well-designed extension services, guidance, and gender equity strengthen women involved in the fishing industry for higher productivity.
- Enrich fishermen with novel technology applications and required skills will be a greater asset for the fishing industry in Sri Lanka.



Gender Issues in SSF



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Invisible Labor in Coastal Fisheries: Gendered Perspectives from Small Scale Fishers in Mullaitivu District

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Abstract

Small-scale fisheries are an imperative livelihood alternative in Mullaitivu. This study intended to identify the role of gender in small-scale fisheries in Mullaitivu with a particular concern for women engaged in the fish value chain. Primary data were gathered through a field survey using simple random sampling technique. A structured questionnaire was administered to a sample of 200 small-scale fishers in the main fishing grounds of Mullaitivu North, Nayaru, and Kokkilai. Qualitative data were recorded through informal discussions with selected respondents. 16% of the sample represented female-headed households while 50% of the male-headed households reported the contribution of female family members in fisheries. Results revealed distinctive gendered roles in small scale fisheries. Men engaged in harvesting operations that demanded more physical strength. Women mainly focused on postharvest stages and pre-harvest stages while women-headed households engaged in active fishing operations. Women received less wages for the same work and had little decision-making power. This is further accentuated by cultural proscriptions, religion, and caste. Undoubtedly, women play a vital role in the fish value chain amidst unpaid household chores. Although the value of the functions performed by women is high, their role in fisheries remains invisible.

Keywords: Gender, small scale fisheries, women

Policy Recommendations:

- Taking steps to inform and sensitize communities on gender issues as means of gender awareness among the fisheries communities.
- Policies to minimize wage discrepancies in fisheries.

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Female hands in Lobster value Chain: A case of South Coast, Sri Lanka

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Abstract

Lobster fisheries in Sri Lanka have been reported as one of the most economically salient and export-oriented industries. Even though, women are playing substantial roles in the sector, there is a limited understanding of women's unique roles and contributions. This study aimed at examining the different types and extent of female participation in lobster value chain. Key areas of gender analysis include decision making on value chain, access to and decision making power over productive resources, control over use of income, skills and capacity, leadership in markets, mobility, self and women's perceptions. Primary data were collected by using interviewer administrated questionnaires and in-depth interviews from 45 lobster value chain actors. Gendered value chain mapping, division of employment, constraint solution tree and win-win strategy tree were main analytical tools. The results revealed that as per the gendered value chain mapping and division of employment, women are playing a hidden role throughout the value chain while no or low level of power in decision making on trading and market access and limited to household activities. Household-level gender inequalities caused to create value chain constraints and according to win-win tree, capacity building, inclusive marketing etc., have to be implemented to break traditionally limited female involvement.

Keywords: Gender, Lobster Fisheries, Sri Lanka

Policy Recommendations:

- Introduction of community-based participatory gender empowerment programs.
- Resource audit, and skill improvement through strategic interventions.

DOI: <https://doi.org/10.6084/m9.figshare.21124882.v1>

**Dissent on women employment in Small-scale Fishing Communities:
A case study at Gandara of Southern Sri Lanka**

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Abstract

In the process of socialization, gender roles in fishing communities make men the breadwinners while women are destined to take care of household responsibilities, raising children and caregiving. The opportunities for women in Small-scale fishing communities are not equal when it comes to employment. In a context of little attention being paid to this fact, the present study was undertaken at Gandara, in Southern Sri Lanka, aiming at identifying the impediments that women encounter in taking up employment. The methodology involved a Focus Group Discussion followed by a questionnaire survey with a sample of 120 respondents. It was revealed that the women's opportunities for engaging in employment are constrained by self-inflicted barriers, with priority being given to bearing household responsibilities, raising children and caregiving. Both parties accepted that there is a certain degree of discrimination against women by not allowing them to engage in employment if the women wish to do so. In resolving this issue, the fisher families proposed to encourage women to take up self-employment activities, that could be carried out at home. Such an approach would minimize the possibility of any negligence of household responsibilities, while making them more financially independent.

Keywords: Gender, Equality, Employment, Barriers

Policy Recommendations:

- Ensure gender equity and equality and promote women employment in fishing communities.

DOI: <https://doi.org/10.6084/m9.figshare.21183865>

**Women contribution in small scale fisheries: special reference to
Kalahagala fishing village in Polonnaruwa district in Sri Lanka**

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Abstract

This research examined how women contribute to fisheries activities, households' consumption, and to total income. Primary data was collected by conducting face-to-face interviews using pre-tested questionnaires from 41.6% of the total population in Kalahagala fishing village located nearby Prakrama Samudraya. Data analysis was performed with descriptive statistics. Women's involvement in fishing is exceptionally high in the sampled village due to easy accessibility to the fishing grounds in the reservoir. Irrespective of race, religion, education, and gender, women members of the fisheries association are involved in fishing alone or with their husbands due to the prevailing economic problems. Fish selling, dry fish production, and repairing and maintaining the fishing gears are the main fishing activities that women performed. About 59% of women engaged in fish selling. Furthermore, 50% of women produced dry fish in the village. The main contribution of women in fisheries is collective harvesting, selling, and dry fish processing which contributes to a 35% share of households' income. In addition, about one to two kilo grams of fish they use for subsistence purposes from the daily catch which is ensuring the nutritional level of the family members. The main drawback for the fishing activities of women is monsoon rain which is disturbed for fishing in the reservoir and processing dry fish.

Keywords: Gender, Freshwater fishery, livelihoods, women contribution

Policy Recommendations:

- Promoting women contribution in fishing is increase the households' income and reduce the women unemployment



Governance and Management of SSF





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Socio-technical barriers Institutional reforms towards transitions of coastal aquaculture in Kerala. Implications for livelihoods and food security

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Abstract

The growth of coastal aquaculture in Kerala, compared to other regions of India, has been slow for a variety of institutional and socio-technical barriers. Recent experiences reveal that there has been a shift in the age-old practice of rice-shrimp rotational cropping system to intensive shrimp culture causing denial of access rights to paddy fields, environmental externalities and livelihood vulnerability. The desire for intensive shrimp monoculture got implanted deeply as salinity incursion in coastal wetlands increased due to climate change. The shift, although increased aquaculture production, raise diseases, poor water quality and escalated social conflicts. Many farmers believe that traditional rice-fish culture might not provide more income to local communities and therefore transition towards monoculture is to be legalised. While the state is still indeterminate on this issue, aquaculture farmers intensified their drive towards mono shrimp culture. The objective of the paper is to document transformations in aquaculture in the Vypin island of Kerala India using an adaptative transformation framework and case study methods. The study indicates that rotational farming would soon vanish as climate change brings in new economic opportunities.

Keywords: Coastal aquaculture, Adaptive transformation, socio-technical networks, Climate change, Institutional reforms

Policy Recommendations:

- Scientific zonation depending on the intensity of salinity incursion is essential for developing climate resilient aquaculture.



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Blue Economy Infrastructure and Small-Scale Fisheries: The Case of Kunthukal Harbour, Pamban Island, Tamil Nadu

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Abstract

Kunthukal harbour is built to encourage deep sea fishing and reduce trawling in the Palk Bay, a major source of conflict between fishers in India and Sri Lanka, it is also part of a Blue Economy vision for harbour-based fisheries. This paper, based on fieldwork amongst small-scale fishers in the study area, analyses the political economy of Blue Economy infrastructure and its impact on small-scale fisheries' livelihoods. The major research questions are: how has the harbour and deep sea fishing affected the coastal ecosystem and the physical and livelihood space for small-scale fisheries?; how have small-scale fishers responded to the construction of the harbour and to the idea of deep sea fishing? This paper utilizes both primary and secondary data; primary data is collected through an extensive interview schedule. Some findings are that small-scale fishers are concerned about the erosion of the seashore at one end and accretion at another end, issues they have raised in public protests. The nature of the seabed has changed after the construction of harbour, the depth of the sea has reduced and fish has declined in the inshore sea. Lastly, this harbour is constructed adjacent to three small-scale fishing hamlets with few if any benefits to them.

Keywords: Blue economy, Small-scale fisheries, Coastal ecosystem, Livelihood

Policy Recommendation-

- A fixed percentage of the budget for the blue economy could be allocated to the development of small-scale fisheries.

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Socio-Economic Struggles of Small-Scale Fishers in Bangladesh

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Abstract

In Bangladesh, the fisheries sector directly employs about 18.5 million people and generates 3.57% of the total GDP and 25.30% of the agricultural GDP. It provides 60% of the daily animal protein intake of Bangladeshis. The goal of this research is to understand the socioeconomic situation of the coastal SSF, to do this primary and secondary data are utilized. Primary data were collected from one coastal district. The sample size was 501 which was calculated using a conventional formula with a 95% confidence level. The research finds that, small-scale fishermen are facing serious socioeconomic issues. The average monthly income of 42.51% of respondents ranges from 6,000 to 15,000 taka. Rohingya influx caused a ban on fishing from a river and thus monthly income of 71.46% SSF fall between 6,000 and 10,000 takas. While the national average monthly income in Bangladesh is 20122 takas. Compared to socioeconomic indicators at the national level, more landless, homeless, powerless, and extremely poor persons are observed among fishers. Concerningly, 62% of women responders said they have experienced various forms of violence. This study provides evidence that SSF households require special consideration. In light of this, the study also makes some specific suggestions.

Keywords: Fisheries, small-scale fishers, socio-economic condition

Policy Recommendations:

- Special schemes for fishers' families
- Access to finance and market for SSF
- Health and education services for SSF
- Access to fishing grounds
- Fisher's participants in decision making
- Special schemes for women



DOI: <https://doi.org/10.6084/m9.figshare.21152089.v1>

Transaction Costs of Community Participation for the Development of Culture-Based Fisheries in Village Reservoirs of Sri Lanka

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Abstract

Fisheries in some small tank cascade systems are managed under different institutional arrangements, viz. farmer organizations, aquaculture committees, renting to a group of fishers. With the recognition of the huge potential of inland reservoirs, culture-based fisheries were developed in seasonal and perennial tanks. The objective of this study is to assess the transaction cost, total revenue and net benefits of fisheries under these three institutional arrangements. The data was collected from 465 fishers and office bearers of the community organizations in four cascades located in Anuradhapura and Kurunegala districts through face-to-face interviews using a structured questionnaire. Transaction costs include costs incurred for gathering information & decision making, implementation of decisions and monitoring. The study finds that renting has the highest transaction cost compared to the other arrangements, which pay more for information gathering and decision making to take a decision for rent and the lowest for fishers in cascades managed under farmer organization. Monitoring costs (approximately 50%) are the most expensive component for fishers in aquaculture committees. Results show that the aquaculture committees generate the highest total revenue and the highest net profit among all other arrangements. This study provides important insights into the cost and benefits associated with different management approaches.

Keywords: Transaction Costs, Fisheries Co-management, Culture-based fisheries, Cascade system

Policy Recommendations:

- The return to fishing is also high in the aquaculture committee system. Thus, to enhance the financial gain from fishing without scarifying environmental sustainability it is recommended to convert the management to the aquaculture committee.





Voice of the People





Threaten Livelihood of SSF

Pakistan's small-scale fishermen have faced many threats and this video depicts their stories. Ibrahim Hydri is the largest fishing settlement on the outskirts of Karachi with a population of approximately 150,000. Eighty per cent of the population of Ibrahim Haidri depends on fishing and associated jobs. Many of them are small-scale fishermen who fish only for their livelihood. Most fishermen had small vessels belonging to the same family or close relatives. Small-scale fishers are facing worse uptown in Pakistan. Climate change primarily affects people's lives and livelihoods, but other factors also apply. But the biggest barrier right now is the high cost of fuel. Many other threats are the large trawlers and harmful nets, marine pollution, deforestation of Mangrove forests, the lack of fresh water in the delta, the rise in oil prices and the fishermen of Pakistan and India for no reason. Rather, to draw attention to the mutual enmity between the two countries, arrest and imprison them for years. While prisoners spend years in Indian prisons, families of fishers are forced to live in extreme hunger and poverty, and children are expelled from school. Karachi city's 550 million gallons of sewage untreated water discharged into the sea, causing severe loss of livelihood of small-scale fishermen mostly fishing in the nearby coastal areas and areas around the Tamar forests. Not only are the roots of the mangrove forests dug by the dirty water of this sewage, but the fish and shrimp nurseries have also been destroyed. Rising prices of oil and ration have made the small fishermen heavily indebted and due to monopoly in the market, the small fishermen do not get even a fair or better rate.

Saheed Baloch

Pakistan Fisherfolk Forum



Livelihood Landscape of Highland Reservoir Fisheries: A Case of Samanalawewa Reservoir of the Central Highlands of Sri Lanka

Samanalawewa reservoir, established primarily for the purpose of hydropower generation and is in Imbulpe DS division, Rathnapura district, Sri-Lanka. It generates hydropower and fosters inland fishing in the limited areas of the highland. Thus, the objective of this study was to explore the livelihood landscape of highland reservoir fisheries. Primary data were collected using the storytelling exercise with the participation of a mix group (10) of fishermen, collectors, and retailers and an in-depth interview with the president of the inland fisheries society. As revealed by the results, a fisheries organization named "Samanalawewa inland fisheries society" has been operating in the reservoir since 2006. The society has 37 registered fishermen with 20 fishing boats. Most importantly, all the fishermen engage in fishing as their secondary source of income, where a fisherman can earn a minimum of 14 USD for 10 kg of the daily catch. Major varieties of fish caught include Tilapia, Catla, and giant freshwater prawns. In particular, giant freshwater prawns from the fishery added colour to the fish basket while entertaining the tourist hotels in the area. The fingerlings are released to the reservoir once every two months during the fishing season. The fishing community is mainly dispersed in the vicinity of the reservoir, especially close to the landing sites; Pitawala-ara, Kinchigune, Karape, Warakawela, and Watawala. The off-fishing season lies between April and September. Owing to the heavy winds in the months of April to September, the fishing season is limited to six months per year. The fisheries society plays a key role in the community by supporting individuals in obtaining fishing gear, boats, and fingerlings. Moreover, the society supports the social development of its members by supporting them in constructing their houses and providing them with land to build a house on their own. The fish catch has a short supply chain where individual fishermen use direct channels to sell the catch to their consumers. Their retail points are their houses or roadside selling points. Fishermen are unable to meet the demand with limited fish supply, and they are exploring avenues to enhance the fish supply or diversify the catch. Only one fisherman has been able to diversify his operation as well as the income through grading, processing (cleaned, gilled, and gutted frozen fish, and packeted fish portions) and retailing through his own channels. Community-based fisheries society's work is male dominant but all work is facilitated by females except fishing. Empowering the society through good postharvest management and management interventions will ensure better returns to the community.

A.M.S.M.R.S.G. Bandara and D.A.M. De Silva

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Small Scale Fishers' and Fish Farmers' Inalienable Right to Water

Small Scale Fishers and Fish Farmers do not have any legal right over the water bodies on which they depend for their livelihood. The informal traditional or customary livelihood practices of fishing and fish farming are encroached upon by competing activities like ports and water transport, tourism, mining and oil exploration, thermal power plants, and also by overfishing and destructive fishing by mechanized fishing fleets. Pollution from industries, agriculture, municipalities, intensive aquaculture, and other activities together with the diversion of water from inland water bodies are wreaking havoc on the ecology, biodiversity, and fish resources associated with water bodies. Even in the case of privately owned ponds and tanks, which are taken on lease by small-scale fish farmers, there is no security of lease. No regulation on fixation or increment of lease rent, and no guarantee of continuation of the lease once the agreement period is over. The small-scale fishers and fish farmers are legal non-entities so far the main resource base of their livelihood is concerned.

The four video clips provide evidence of these issues

1. Overfishing by Trawlers and Mechanised Boats and hazards of tourism; [1.18 minute]
2. Reduction of fish resources due to pollution, and construction of breakwaters. Hazards of fishing due to river traffic, national waterway [1.22 minute]
3. Hazards of intensive shrimp farming [1.22 minute]
4. Trawlers damage nets, overfishing and destructive fishing, and need for infrastructure like sheds, drinking water, toilets, etc. Need for quality fish farming inputs in a cheap rate. [1.36]

Pradip Chatterjee

National Platform for Small Scale Fish Workers (NPSSFW) of India



Let's protect the fishing industry and secure our future dreams

Let us protect our fisher industry... Let us save our dream for tomorrow

Canoe trunk broken, oar is cracked

Lost everything which was gathered

The heap of memories of our foot prints

Got washed away on the shore today

The happy childhood memories of smiling faces along this shore
Has been shattered with the erection of high rising palaces of powerful

Though we were a free and independent people

Today all of us, entire fisher people have become mere labourers

Eyes of the fisher mothers and fathers become blurred

With the development plans of top class

The small people like us has been pushed to the border

Serenity of the Ocean has been destroyed

With the mounting of illegal fishing nets and dynamites

Our hearts are grieving and minds have become numb

Seeing what is happening in front of our eyes

It is enough now keeping silent

Let us protect our fishing industry

Let us clam our and raise our voices to nourish the fish supply

Now it is the right time

Let us all be awake

Let us get together all small fishers

Come forward to display our power to the authorities

Let us protect our fisher industry.....

Let us save our dream for tomorrow.....

Let us protect our fisher industry.....

Let us save our dream for tomorrow.....

Francis Priyankara Costa

Praja Abhilasha network/NAFSO Sri Lanka

A Healthy and a Resilient Lagoon

Negombo lagoon is a natural water entity situated around 40 Km north of Colombo and surrounded by the Indian Ocean at the Negombo coastal belt. It has been declared as a very sensitive and fertile zone by the scientists of the National Aquatic Resources and Development Agency (NARA). It consists of 13 small islands closer to Negombo with very rare beauty and aesthetic touch. The largest of all islands of this lagoon belt, namely Munnakakraya and 4 other isles has been populated and the rest has become a homeland for a fertile mangrove. The primary inflows of clean water have been from the Attanagalu Oya (Dadugam Oya), Jaela canal, and Hamilton canal (Old Dutch Canal). It is not an exaggeration if we are to describe this mangrove zone as the source of Oxygen provider to the urban population residing in the belt of Negombo to Colombo. The lagoon has extensive mangrove swamps (around 29 different varieties of mangroves) which attracts a variety of water birds including *cormorants, herons, egrets, gulls, terns*, and other various types. It also has become a breeding ground for *prawns and crabs* and around 140 different types of fish. Around 6000 fisher families reside along the Lagoon belt and make their living from this water resource. Negombo lagoon, which is linked to the sea (Indian Ocean) by a narrow strip to the north, is one of the largest estuarine lagoons in Sri Lanka, situated in the southwest part, with a surface area of 35.02km².

This rich natural water body is in danger today due to a number of careless human interventions and imprudent implementation of accelerated development plans. The lagoon is being choked to death by the:

- *Increased dumping of Garbage & Waste to the Lagoon*
- *Substantial contribution by Dry Fish Processing Units towards the Pollution*
- *Gathering of Plastic waste instead of moss and other aquatic plants*
- *Improper implementation of the Negombo Lagoon Development Plan.*

The proposed Negombo Lagoon Development Plan is not being implemented towards the desired direction of reviving the life and quality of Lagoon. Hence our efforts on raising awareness, organizing and empowering fishing communities around the lagoon and get them mobilized and engaged in the process of healing and protecting the Lagoon.

Nimal I Perera

Janwaboda Kendraya, Negombo



Challenges Faced by Fisherwomen

Women in the fishing industry account for at least 50% of the progress of the fishing industry. Fishing was a family enterprise at the time of traditional fishing, as men and women had an understanding of their role in fishing, other social activities, and family. Women engaged in various aspects of the fishing value chain from pre-harvest to postharvest practices as collecting fish from the net, sorting out the quality, selling on the spot, or engaging as fish vendors, fish drying, salting, etc. Although women are primarily responsible for managing family income, they must take care of their husbands, particularly when there is no income. However, the role of women in society is poorly recognized. In the post-technological era, the influx of capital in the industry caused the elimination of women from fishing activities. As multi-day boat operations are primarily based in ports and the participation of women becomes less significant. Even in the SSF, it is unfortunate that women's involvement is not visible and not properly recognized today. Women have no place in fishing cooperatives. The women who became the breadwinner of the family after losing their husbands during the war did not pay enough attention to making a decent living. There should be some mechanism to uplift the living condition of those women and their children's education and safety. Fishing cooperatives are the appropriate structures to ensure their security, support livelihoods, and meet essential needs.

We expect the authorities to give sufficient attention to the needs of women in the fishery and take the necessary steps to build fishing communities during the IYAF.

Presenter- Subhashini Deepa

Sri Wimukthi Fisher Women Organization of Sri Lanka





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